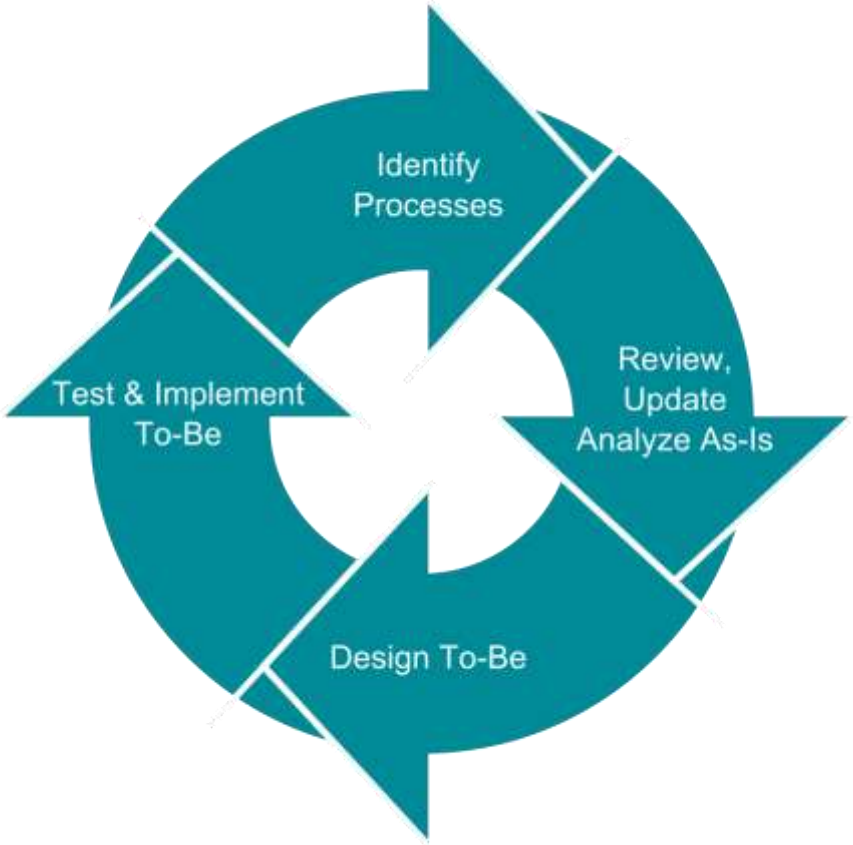
Business Process Engineering
Week # 13

https://www.lovelycoding.org/banking-management-system/

Reengineering

- "Reengineering is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical, contemporary measures of performance, such as cost, quality, service, and speed."

Business process reengineering is also known as business process redesign, business transformation, or business process change management.



Business Process Reengineering Cycle

Business process redesign

- Goal is to rethink and streamline business processes
- aim is major gains in cost, quality, time-to-market, etc.
- also called business process reengineering or business reengineering or just reengineering

Reengineering

- Essence of reengineering - discontinuous thinking - break away from outdated rules and assumptions that underlie operations
- Reengineering requires looking at fundamental processes of business from a cross-functional perspective

Reengineering guidance and relationship of mission and work processes to information technology.



Outdated business processes

- stop embedding outdated processes in computerized systems
- companies tend to use technology to mechanize old ways of doing business
- business processes came of age in different competitive environment before advent of computer

Outdated business processes

- much work organized as sequence of separate tasks
- use complex mechanisms to track its progress

Outdated business processes

- conventional process structures are fragmented and lack integration necessary to maintain quality and service
- people tend to substitute the narrow goals of their department for larger goals of process as a whole

BUSINESS PROCESS REENGINEERING

- The search for, and implementation of, radical change in business processes to achieve breakthrough results
- Synonyms: business process redesign, business transformation, process innovation, business reinvention, change integration
- Starting point: clean sheet of paper

BUSINESS PROCESS REENGINEERING

- Start from the future and work backwards
- BPR is not easy - serious work
- BPR is not free - financial & cultural
- BPR often driven by fear and greed
- Change is a “struggle”; BPR is a “war”

BPR: LESSONS LEARNED

- Get the strategy straight first
- Lead from the top
- Create a sense of urgency
- Design from the outside in (customer)
- Combine top-down & bottom-up efforts

BPR: STRATEGY

- Stakeholder Assessment - shareholders, customers, employees
- Determine which stakeholder expectations should be met to gain competitive advantage
- Determine how to redesign to meet expectations
- Map out IT solutions to support
- Develop & implement new processes, etc.

IT to the rescue

- use IT to radically redesign business processes in order to achieve dramatic improvements in performance
- examples: Ford and MBL (mutual benefit life ins.)

Principles of Reengineering

- Organize around outcomes, not tasks
- Consider those who use the output of the process perform the process

Principles of Reengineering

- Put the decision point where the work is performed, and build control into the process
  - Old Perception-hierarchical management structure built on the assumption that people who actually do the work cannot/will not monitor and control it and that they lack knowledge and scope to make decisions about it
  - new principle - people who do work should make decisions and that process itself can have built-in controls

Risky business

- what are risks associated with reengineering?

Change

- Reengineering necessitates huge organizational change
- no-one likes change
- so need executive leadership and vision
- very stressful to implement a reengineering plan

Uncertain Results

- reengineering cannot be planned and accomplished in small and cautious steps
- - it's an all or nothing proposition with an uncertain result

Organizational Resistance

- Reengineering teams find that making progress in early organizing and creative analysis stages is easy and exciting.

- Later work of development and implementation is much more difficult -- organizational resistance

Many failures

- one big reason for failure is due to not making IS a real partner in reengineering
  - Not Getting Buy-In From The Company
  - Not Putting The Right Team Together
  - Inability to Properly Analyze Processes

TYPICAL KPIs

- Vary depending on what process you're optimizing, the following can be very typical:

- Manufacturing
  - Cycle Time – The time spent from the beginning to the end of a process
  - Changeover Time – Time needed to switch the line from making one product to the next
  - Defect Rate – Percentage of products manufactured defective
  - Inventory Turnover – How long it takes for the manufacturing line to turn inventory into products
  - Planned VS Emergency Maintenance – The ratio of the times planned maintenance and emergency maintenance happen

- IT
  - Mean Time to Repair – Average time needed to repair the system / software / app after an emergency
  - Support Ticket Closure rate – Number of support tickets closed by the support team divided by the number opened
  - Application Dev. – The time needed to fully develop a new application from scratch
  - Cycle Time – The time needed to get the network back up after a security breach

What is role of IT in reengineering?

- IT is an essential enabler in reengineering - supports redesigned business processes and facilitates cross-functional work flow

SOFTWARE RE-ENGINEERING

- RESTRUCTURING - Transforming unstructured (“spaghetti code”) code into structured code equivalent [10% penetration]
- RE-ENGINEERING - Gradually “clean up” old software without affecting user functionality
- REVERSE ENGINEERING - Uncover or rediscover the design specification from existing code; then apply forward engineering